

AdPulse - Project Guide

AI-Powered Ad Campaign Performance Analyzer | MSIS 521 | University of Washington | Winter 2025

Part 1: Detailed Technical Explanation

What We Built

AdPulse is a web-based tool that helps marketers understand **why some ads perform better than others**. It takes real advertising data (247 Super Bowl ads, 2000-2020) and runs AI and statistical analyses, presenting results through an interactive Streamlit dashboard. The tool is extensible: anyone can upload their own campaign CSV and get the same analyses on any set of ads.

The Data

We used a real dataset from **FiveThirtyEight**, enhanced by **TidyTuesday**. FiveThirtyEight staff watched every Super Bowl ad (2000-2020) and tagged each with characteristics (funny, celebrity, patriotic, danger, animals, use_sex, show_product_quickly). The TidyTuesday version added **YouTube engagement data** (views, likes, dislikes, comments, favorites) plus video titles and descriptions. This gives us **247 ads across 10 major brands** (Bud Light, Budweiser, Coca-Cola, Doritos, Hyundai, Kia, NFL, Pepsi, Toyota, E-Trade) with **25 columns** combining human-coded characteristics and real engagement numbers.

The Six Core Analyses

1. Campaign Dashboard

Overview page with four KPIs (total ads, total views, average views, brand count), a views-over-time line chart, and brand comparison. Built with pandas aggregation and Plotly interactive charts. Streamlit sidebar provides year-range and brand filters across all pages.

2. Ad Performance Ranking

A **composite engagement score** (0-100) weights multiple metrics: Views 40%, Likes 35%, Comments 25%. Each metric is min-max normalized to 0-1, multiplied by its weight, summed, then scaled to 100. This single number balances reach with active audience engagement for fair cross-ad comparison.

3. Feature Impact Analysis

For each of the 7 boolean features, we split ads into WITH vs. WITHOUT groups and compute: $\text{views_lift} = (\text{avg_with} / \text{avg_without} - 1) \times 100$. **Key finding:** "Show Product Quickly" gives +70% lift; celebrity appearances show -54% lift (counterintuitive but real in this dataset).

4. Sentiment Analysis

We use **VADER** (Valence Aware Dictionary and sEntiment Reasoner) from NLTK on ad descriptions. VADER assigns compound scores (-1 to +1); we classify ≥ 0.05 as positive, ≤ -0.05 as negative, rest neutral. An optional BERT transformer model is available for higher accuracy. **Key finding:** ~75% of descriptions are positive; brands deliberately craft optimistic narratives.

5. Topic Modeling (LDA)

We use **Latent Dirichlet Allocation** to discover themes across 207 descriptions. The pipeline: (a) preprocess text (remove URLs/punctuation, lowercase, tokenize, remove stopwords, lemmatize); (b) vectorize with CountVectorizer (max_df=0.95, min_df=2, max 1000 features); (c) fit LDA with 5 topics, extracting top 10 words per topic; (d) assign each ad its dominant topic. **Key finding:** Topics cluster around brands (Hyundai, Bud Light, Doritos) and themes (NFL/football).

6. Search Trends

Integrates **Google Trends** via pytrends to show how brand search interest changes around Super Bowl time, serving as a proxy for ad effectiveness beyond YouTube views.

Output Validation

Sentiment Distribution Check: Flags results where a single class exceeds 95% (suggests a broken model). **Known-Sample Accuracy:** Runs VADER on 5 hand-picked test sentences with known answers and reports accuracy (e.g., 4/5 = 80%). **Topic Distinctness Score:** Computes pairwise cosine similarity between topic-word distributions; $\text{distinctness} = 1 - \text{avg_similarity}$. Higher means more distinct topics; alerts if LDA produces redundant output.

Technology Stack and File Structure

Layer	Tech	Purpose
UI	Streamlit	Interactive web dashboard
Data	pandas, NumPy	Data manipulation
NLP	NLTK (VADER)	Sentiment analysis, tokenization
ML	scikit-learn	LDA topic modeling
Charts	Plotly	Interactive visualizations
Trends	pytrends	Google Trends API

File	Purpose
app.py	Streamlit dashboard (7 pages, ~700 lines)
analysis.py	Sentiment, topics, engagement, validation (~240 lines)
data_utils.py	Data loading, cleaning, NLP preprocessing (~114 lines)
fetch_comments.py	YouTube comment fetcher (~150 lines)
create_deck.py	Generates 12-slide .pptx from data (~660 lines)
data/superbowl_ads.csv	TidyTuesday dataset (247 rows, 25 cols)

Part 2: The Layman's Explanation

This section explains the project in plain English for anyone new to marketing analytics or AI.

What is the big picture?

Imagine you are a marketing manager at Doritos. You just spent **\$7 million** on a 30-second Super Bowl ad. Your boss asks: "Did it work? Should we do the same thing next year?" Traditionally, you would look at TV ratings ("113 million people watched") or run a focus group. But that is vague. Did people actually *engage* with your ad? Did they *like* it? What did they *talk about*? Was it the humor that worked, or showing the chips in the first 5 seconds? **AdPulse answers these questions using data and AI.**

How does it work?

Think of AdPulse as doing five things a human analyst would do, but faster and at scale:

1. "How popular was each ad?" - Engagement Scoring

YouTube tells us views, likes, and comments for each ad, but raw numbers are hard to compare. An ad with 5 million views and 100,000 likes is not obviously better than one with 2 million views and 200,000 likes. So we create a **single score from 0 to 100** that blends all three metrics. Think of it like a GPA: instead of individual grades, you get one number summarizing overall performance.

2. "What makes an ad work?" - Feature Impact

Every ad was tagged by reviewers: funny, celebrity, patriotic, shows product quickly, etc. We split all 247 ads into groups (e.g., funny vs. not funny) and compare average views. The surprising answer: **showing your product quickly** matters far more than hiring a celebrity. Ads that show the product early get 70% more views. Celebrity ads actually get *fewer* views on average.

3. "What is the vibe?" - Sentiment Analysis

We feed each ad's text description into **VADER**, a program that reads words and determines the overall tone (positive, negative, or neutral). How? Imagine a huge dictionary where every word has a "mood score": "amazing" scores positive, "terrible" scores negative, "table" is neutral. VADER reads all words, accounts for negation ("NOT amazing") and intensifiers ("VERY amazing"), and outputs an overall score. It is not perfect (it does not truly *understand* language like a human), which is why we include sanity checks.

4. "What are people talking about?" - Topic Discovery

Reading 207 ad descriptions and summarizing themes manually would take hours. **LDA (Latent Dirichlet Allocation)** does this automatically. Think of it like sorting newspaper articles into piles by topic. You do not tell the algorithm what the topics are; it figures them out by looking at which words appear together. Sports articles cluster because they share "game," "team," "score." LDA found 5 themes: Hyundai campaigns, Bud Light/Budweiser, recent ads, Doritos contests, and NFL/football.

5. "Can we trust these results?" - Output Validation

AI can give confident-sounding but wrong answers, so we built in three automatic checks:

- **Distribution check:** If the tool says 98% of ads are "positive," something is likely broken. We flag any result where one category dominates too heavily (>95%).
- **Known-answer test:** We run the tool on sentences where we already know the answer ("This is amazing" = positive, "This is terrible" = negative). If it gets those wrong, we know something is off.
- **Topic overlap check:** If all 5 "topics" contain the same words, they are not really different. We measure how distinct each topic is from the others.

Why does this matter?

Marketing has traditionally been a gut-feeling industry: "I *think* humor works," "I *feel* like we need a celebrity." AdPulse brings **data to the conversation**. Instead of guessing, you can see that across 20 years of Super Bowl data, showing your product quickly consistently outperforms celebrity appearances. And because the tool accepts any CSV of ad data, it is not limited to Super Bowl. A startup analyzing Instagram campaigns could upload their data and get the same analyses.

The Presentation

We also generated a 12-slide PowerPoint deck summarizing the problem, solution, data, technical approach, key findings, and output validation. It is designed for a 15-minute class presentation with a live demo of the interactive dashboard.